In [ ]:
```
# Clean up and install compatible versions of torchao and diffusers
!pip uninstall -y torchao diffusers transformers
!pip install "torchao<0.16.0" "diffusers>=0.31.0" transformers accelerate
```

```
Found existing installation: torchao 0.10.0
Uninstalling torchao-0.10.0:
  Successfully uninstalled torchao-0.10.0
Found existing installation: diffusers 0.41.0
Uninstalling diffusers-0.41.0:
  Successfully uninstalled diffusers-0.41.0
Found existing installation: transformers 5.20.0
Uninstalling transformers-5.20.0:
  Successfully uninstalled transformers-5.20.0
Collecting torchao<0.16.0
  Downloading torchao-0.15.0-cp310-abi3-manylinux_2_24_x86_64.manylinu
x_2_28_x86_64.whl.metadata (22 kB)
Collecting diffusers>=0.31.0
  Using cached diffusers-0.41.0-py3-none-any.whl.metadata (20 kB)
Collecting transformers
  Using cached transformers-5.20.0-py3-none-any.whl.metadata (31 kB)
Requirement already satisfied: accelerate in /usr/local/lib/python3.13/di
st-packages (1.15.0)
Requirement already satisfied: safetensors in /usr/local/lib/python3.13/d
ist-packages (0.8.0)
Requirement already satisfied: torchvision in /usr/local/lib/python3.13/d
ist-packages (0.29.1)
Requirement already satisfied: importlib_metadata in /usr/local/lib/pytho
n3.13/dist-packages (from diffusers>=0.31.0) (9.0.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.13/dis
t-packages (from diffusers>=0.31.0) (4.0.7)
Requirement already satisfied: huggingface-hub<2.0,>=1.32.0 in /usr/loca
l/lib/python3.13/dist-packages (from diffusers>=0.31.0) (1.33.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-pa
ckages (from diffusers>=0.31.0) (2.1.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python
3.13/dist-packages (from diffusers>=0.31.0) (2025.11.3)
Requirement already satisfied: requests in /usr/local/lib/python3.13/dis
t-packages (from diffusers>=0.31.0) (2.32.4)
Requirement already satisfied: Pillow in /usr/local/lib/python3.13/dist-p
ackages (from diffusers>=0.31.0) (11.3.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python
3.13/dist-packages (from transformers) (26.3)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.13/d
ist-packages (from transformers) (6.0.3)
Requirement already satisfied: tokenizers<0.24.0,>=0.23.1 in /usr/local/l
ib/python3.13/dist-packages (from transformers) (0.23.2)
Requirement already satisfied: typer in /usr/local/lib/python3.13/dist-pa
ckages (from transformers) (0.27.2)
Requirement already satisfied: tqdm>=4.60 in /usr/local/lib/python3.13/di
st-packages (from transformers) (4.67.3)
Requirement already satisfied: psutil in /usr/local/lib/python3.13/dist-p
ackages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.13/
dist-packages (from accelerate) (2.14.1)
Requirement already satisfied: click<9.0.0,>=8.4.2 in /usr/local/lib/pyth
on3.13/dist-packages (from huggingface-hub<2.0,>=1.32.0->diffusers>=0.3
1.0) (8.5.0)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python
3.13/dist-packages (from huggingface-hub<2.0,>=1.32.0->diffusers>=0.31.0)
```

```
(2025.12.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.6.0 in /usr/local/lib/pyth
hon3.13/dist-packages (from huggingface-hub<2.0,>=1.32.0->diffusers>=0.3
1.0) (1.6.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python
3.13/dist-packages (from huggingface-hub<2.0,>=1.32.0->diffusers>=0.31.0)
(0.28.1)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/li
b/python3.13/dist-packages (from huggingface-hub<2.0,>=1.32.0->diffuser
s>=0.31.0) (4.16.0)
Requirement already satisfied: setuptools>=77.0.3 in /usr/local/lib/pytho
n3.13/dist-packages (from torch>=2.0.0->accelerate) (80.10.2)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.1
3/dist-packages (from torch>=2.0.0->accelerate) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python
3.13/dist-packages (from torch>=2.0.0->accelerate) (3.7)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-p
ackages (from torch>=2.0.0->accelerate) (3.1.6)
Requirement already satisfied: cuda-toolkit==13.0.3 in /usr/local/lib/pyt
hon3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupt
i,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_system
== "Linux"->torch>=2.0.0->accelerate) (13.0.3.0)
Requirement already satisfied: cuda-bindings<14,>=13.0.3 in /usr/local/li
b/python3.13/dist-packages (from torch>=2.0.0->accelerate) (13.0.3)
Requirement already satisfied: nvidia-cudnn-cu13==9.24.0.43 in /usr/loca
l/lib/python3.13/dist-packages (from torch>=2.0.0->accelerate) (9.24.0.4
3)
Requirement already satisfied: nvidia-cusparselt-cu13==0.8.1 in /usr/loca
l/lib/python3.13/dist-packages (from torch>=2.0.0->accelerate) (0.8.1)
Requirement already satisfied: nvidia-nccl-cu13==2.30.7 in /usr/local/li
b/python3.13/dist-packages (from torch>=2.0.0->accelerate) (2.30.7)
Requirement already satisfied: nvidia-nvshmem-cu13==3.4.5 in /usr/local/l
ib/python3.13/dist-packages (from torch>=2.0.0->accelerate) (3.4.5)
Requirement already satisfied: triton~=3.8.0 in /usr/local/lib/python3.1
3/dist-packages (from torch>=2.0.0->accelerate) (3.8.0)
Requirement already satisfied: nvidia-cublas==13.1.1.3.* in /usr/local/li
b/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufil
e,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platfor
m_system == "Linux"->torch>=2.0.0->accelerate) (13.1.1.3)
Requirement already satisfied: nvidia-cuda-nvrtc==13.0.88.* in /usr/loca
l/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cuf
ile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platfor
m_system == "Linux"->torch>=2.0.0->accelerate) (13.0.88)
Requirement already satisfied: nvidia-cuda-runtime==13.0.96.* in /usr/loc
al/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cu
file,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platfo
rm_system == "Linux"->torch>=2.0.0->accelerate) (13.0.96)
Requirement already satisfied: nvidia-cufft==12.0.0.61.* in /usr/local/li
b/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufil
e,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platfor
m_system == "Linux"->torch>=2.0.0->accelerate) (12.0.0.61)
Requirement already satisfied: nvidia-nvjitlink<14,>=13.0.88 in /usr/loca
l/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cuf
ile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platfor
m_system == "Linux"->torch>=2.0.0->accelerate) (13.0.88)
```

```
Requirement already satisfied: nvidia-cufile==1.15.1.6.* in /usr/local/li
b/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufil
e,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platfor
m_system == "Linux"->torch>=2.0.0->accelerate) (1.15.1.6)
Requirement already satisfied: nvidia-cuda-cupti==13.0.85.* in /usr/loca
l/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cuf
ile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platfor
m_system == "Linux"->torch>=2.0.0->accelerate) (13.0.85)
Requirement already satisfied: nvidia-curand==10.4.0.35.* in /usr/local/l
ib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufil
e,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platfor
m_system == "Linux"->torch>=2.0.0->accelerate) (10.4.0.35)
Requirement already satisfied: nvidia-cusolver==12.0.4.66.* in /usr/loca
l/lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cuf
ile,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platfor
m_system == "Linux"->torch>=2.0.0->accelerate) (12.0.4.66)
Requirement already satisfied: nvidia-cusparse==12.6.3.3.* in /usr/local/
lib/python3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufil
e,cupti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platfor
m_system == "Linux"->torch>=2.0.0->accelerate) (12.6.3.3)
Requirement already satisfied: nvidia-nvtx==13.0.85.* in /usr/local/lib/p
ython3.13/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cup
ti,curand,cusolver,cusparse,nvjitlink,nvrtc,nvtx]==13.0.3; platform_syste
m == "Linux"->torch>=2.0.0->accelerate) (13.0.85)
Requirement already satisfied: zipp>=3.20 in /usr/local/lib/python3.13/di
st-packages (from importlib_metadata->diffusers>=0.31.0) (4.1.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/li
b/python3.13/dist-packages (from requests->diffusers>=0.31.0) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/
dist-packages (from requests->diffusers>=0.31.0) (3.20)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/pytho
n3.13/dist-packages (from requests->diffusers>=0.31.0) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/pytho
n3.13/dist-packages (from requests->diffusers>=0.31.0) (2026.7.22)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/pytho
n3.13/dist-packages (from typer->transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.13/
dist-packages (from typer->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/pyt
hon3.13/dist-packages (from typer->transformers) (0.0.5)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/pyt
hon3.13/dist-packages (from cuda-bindings<14,>=13.0.3->torch>=2.0.0->acce
lerate) (1.8.2)
Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-pa
ckages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.32.0->diffuser
s>=0.31.0) (4.15.1)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.1
3/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.32.0->dif
fusers>=0.31.0) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.13/dis
t-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hu
b<2.0,>=1.32.0->diffusers>=0.31.0) (0.16.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/py
thon3.13/dist-packages (from rich>=13.8.0->typer->transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/
```

```
python3.13/dist-packages (from rich>=13.8.0->typer->transformers) (2.2
1.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/pytho
n3.13/dist-packages (from sympy>=1.13.3->torch>=2.0.0->accelerate)
(1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python
3.13/dist-packages (from jinja2->torch>=2.0.0->accelerate) (3.0.3)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.13/di
st-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformer
s) (0.1.2)
Downloading torchao-0.15.0-cp310-abi3-manylinux_2_24_x86_64.manylinux_2_2
8_x86_64.whl (7.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 7.2/7.2 MB 88.5 MB/s eta 0:0
0:00
Using cached diffusers-0.41.0-py3-none-any.whl (6.1 MB)
Using cached transformers-5.20.0-py3-none-any.whl (12.8 MB)
Installing collected packages: torchao, diffusers, transformers
Successfully installed diffusers-0.41.0 torchao-0.15.0 transformers-5.2
0.0
```

In [ ]:
```python
# Import PyTorch and necessary modules for image processing and pipelines
import torch
from diffusers import StableDiffusionImg2ImgPipeline
from PIL import Image
import requests
from io import BytesIO

# Set device to GPU (cuda) for fast generation
device = "cuda" if torch.cuda.is_available() else "cpu"
print(f"Using device: {device}")
```

```
WARNING:torchao:Skipping import of cpp extensions due to incompatible tor
ch version 2.14.1+cu130 for torchao version 0.15.0             Please see
https://github.com/pytorch/ao/issues/2919 for more info
Using device: cuda
```

In [ ]:
```python
# Define a stable pre-trained model capable of image-to-image translation
model_id = "runwayml/stable-diffusion-v1-5"

# Load the image-to-image pipeline with float16 optimization for Colab's
pipe = StableDiffusionImg2ImgPipeline.from_pretrained(
    model_id,
    torch_dtype=torch.float16
)

# Move the pipeline to the GPU
pipe = pipe.to(device)
print("Image-to-Image pipeline loaded successfully!")
```

```
/usr/local/lib/python3.13/dist-packages/diffusers/utils/deprecation_util
s.py:23: FutureWarning: `torch_dtype` is deprecated and will be removed i
n version 1.0.0. Please use `dtype` instead.
  deprecate("torch_dtype", "1.0.0", _TORCH_DTYPE_DEPRECATION_MESSAGE)
Warning: You are sending unauthenticated requests to the HF Hub. Please s
et a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthentica
ted requests to the HF Hub. Please set a HF_TOKEN to enable higher rate l
imits and faster downloads.
```

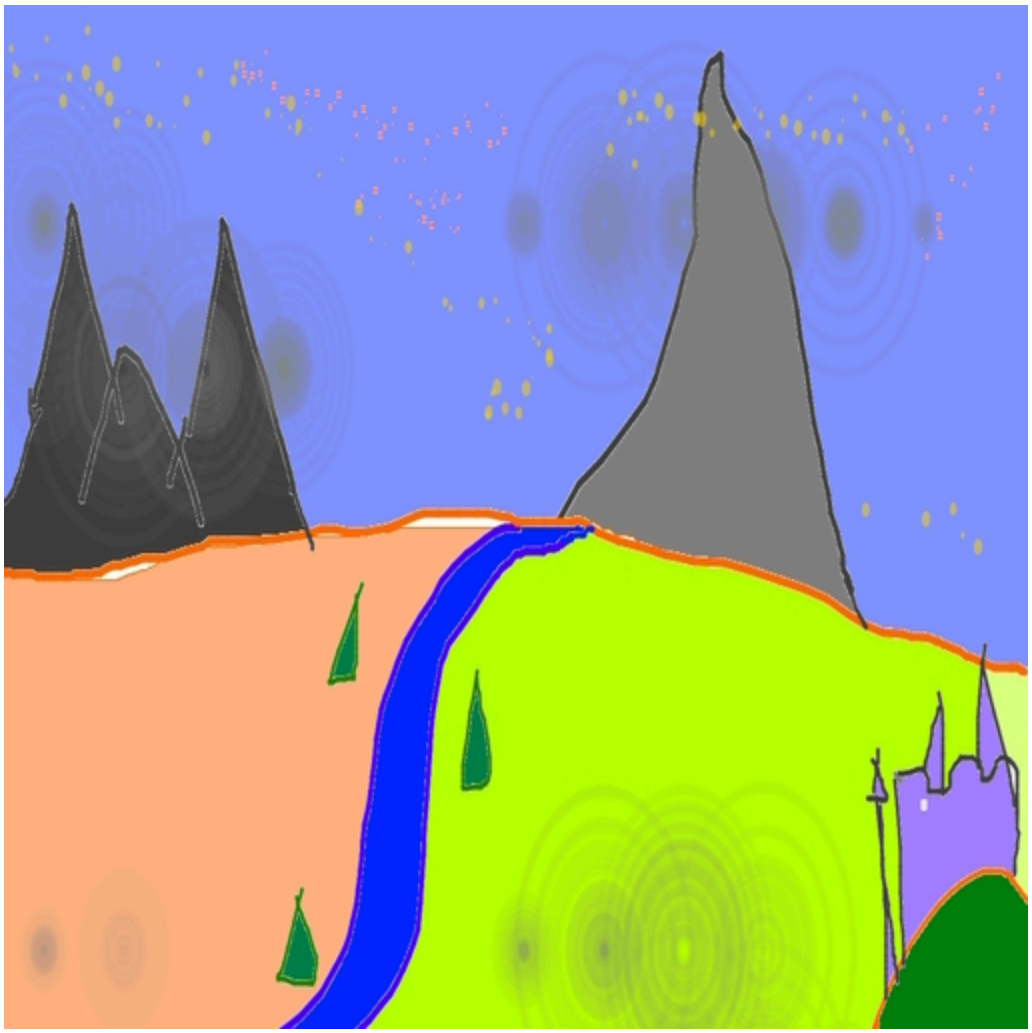
```
Image-to-Image pipeline loaded successfully!
```

In [ ]:
```python
# Download a sample image to use as input for image-to-image translation
url = "https://raw.githubusercontent.com/CompVis/stable-diffusion/main/as
response = requests.get(url)
init_image = Image.open(BytesIO(response.content)).convert("RGB")

# Resize the image to 512x512 pixels for the model
init_image = init_image.resize((512, 512))

# Display the input image in Colab
print("Input Image (e.g., a simple sketch):")
display(init_image)
```

```
Input Image (e.g., a simple sketch):
```



In [ ]:
```
# Write a prompt describing how you want to transform the input image
prompt = "A fantasy landscape with vibrant colors, highly detailed, profe

# Perform image-to-image translation using the model
# strength controls how much the model changes the original image (0.0 to
images = pipe(prompt=prompt, image=init_image, strength=0.75, guidance_sc

# Get the first generated output image
result_image = images[0]

# Display the translated output image
print("Translated Output Image:")
display(result_image)
```

```
Translated Output Image:
```



In [ ]: